Saurav Kumar Verma

ee25mtech14008@iith.ac.in | skvermadbg4@gmail.com | +91 7543861026 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.19
Jesus & Mary Academy	Class XII (CBSE)	2017	81.80%
Delhi Public School	Class X (CBSE)	2015	9.00%

SKILLS

Hardware Description Languages : Verilog, SystemVerilog (basics)

Tools : Cadence Virtuoso,Xilinx Vivado,LT-Spice,MATLAB

Programming & Scripting Language : C , C++, Python

RELEVANT COURSES

Digital Domain : Digital IC Design, CMOS Digital Design, Introduction to HDL, Digital IC design lab.

Analog Domain : Analog IC Design,Analog IC Design Lab.

Devices Domain : Semiconductor Device Modelling, Mesoscopic Device Electronics,Introduction to VLSI Design, Nanoelectronics.

Memory Domain : Neuromorphic Devices for AI/ML,Memory Circuits and System Design (**By Micron).**

MTECH THESIS

Title : Image Sensor - Signal Processor Interfacing.

Description:

- Acquired image data from the ROIC (Readout Integrated Circuit) and implemented high-throughput video transmission using a frame grabber.
- Enabled 24-bit parallel image data transfer via the Camera Link protocol over LVDS physical-layer communication.

PROJECTS

RTL Design of 32-bit MIPS Processor

- Implemented a 32-bit MIPS processor with a 5-stage pipeline architecture, comprising of fetch, decode, execute, memory and write-back stages.
- Implemented the design in Verilog and verified the functionality in Vivado.

AMBA APB Bus Design & Verification using system verilog

- Designed behavioral RTL model of AMBA APB protocol using Verilog.
- Implemented SystemVerilog verification testbench with reusable, class-based architecture (generator, driver, monitor,
- environment, scoreboard).
- Verified correctness by comparing DUT outputs with expected results across test iterations

Dual Clock Asynchronous FIFO:

- Designed a dual-clock asynchronous FIFO in Verilog HDL with independent read/write clocks.
- Implemented two-stage synchronizers with Gray-code pointers for reliable clock-domain crossing.
- Added extra pointer bit to accurately detect full and empty conditions.

Traffic Light Controller -- FPGA Implementation & RTL Verification | Verilog, Vivado, ZedBoard

- Designed a synthesizable FSM-based traffic light controller in Verilog with defined state transitions and timing control.
- Developed testbenches to verify state transitions and corner cases, with waveform-based debugging using Vivado.
- Implemented the complete RTL design on ZedBoard FPGA and validated real-time hardware operation using LED indicators.

Design and Verification of a 4x4 DRAM Array Read-Path

- Designed a transistor-level 4x4 DRAM read-path in Cadence Virtuoso, integrating 1T-1C cells, 2-to-4 row decoders, and latch-type sense amplifiers.
- Implemented cross-coupled level shifters and VDD/2 bitline equalization to overcome NMOS threshold drops and maximize sensing margins.
- Analyzed charge-sharing transients and optimized timing controls to reliably amplify low-differential signals and guarantee complete data write-back.

2-Stage Operational Transconductance Amplifier (OTA)

- Designed and optimised a two-stage Miller-compensated OTA with single-ended output in TSMC 180 nm CMOS technology.
- Achieved 63.54° phase margin, 5.9 V/µs slew rate, and 0.2-1.08 V ICMR at 1.8 V supply operation.

PROFESSIONAL WORK EXPERIENCE

Software Engineer - BreachLock | Nov 2023 - Sep 2024

- Developed onboarding and licensing modules to streamline user and organizational workflows.
- Built backend services using Python and FastAPI, with frontend components using React.js, Angular.js and JavaScript.
- Developed an activity monitoring module for tracking user actions and compliance.
- Automated generation of PDF reports for vulnerability scan results and risk mitigation.

Associate Software Engineer - Accenture | Oct 2022 - Oct 2023

- Worked on application development and enhancement across multiple client projects.
- Developed and integrated application components using Java, React.js and JavaScript.
- Contributed to feature development, debugging and maintenance of enterprise applications.
- Worked with cross-functional teams to deliver application improvements and resolve technical issues.

POSITION OF RESPONSIBILITY

Lead Placement Coordinator - Electrical Engineering Department, OCS IIT Hyderabad

Teaching Assistant, CMOS Digital Design -- under Prof. Dr Kapil Jainwal

EXTRACURRICULAR

Chess, Badminton, Cooking, Strength Training